ASSIGNMENT 1

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Roll no:13155

#import libraries

```
import pandas as pd
import numpy as np
```

#import the dataset

```
df=pd.read_csv(r"C:\Users\System21\Downloads\data.csv")
df
```

	Unnamed: 0	A	B	C	D
0	0	-1.557181	0.373416	1.040685	-0.648513
1	1	-0.835872	-0.005259	0.502099	-0.956839
2	2	-0.257153	2.638228	-0.535731	-1.069443
3	3	-0.505502	-0.339260	0.602622	-1.420995
4	4	1.366398	-0.368296	0.642091	1.609680
5	5	-1.350101	0.210097	-0.743280	0.035729
6	6	-0.621305	-0.620733	0.141429	-1.831875
7	7	-0.670370	-0.379230	1.950228	-0.487202
8	8	-1.360577	0.044675	-0.576289	0.998686
9	9	-0.120302	-0.146491	-0.420956	1.412786
10	10	0.216936	0.398267	0.342519	0.387263
11	11	0.567832	-2.023668	-0.315318	-0.064349
12	12	-0.161148	-0.118924	-1.226679	-0.932540
13	13	-1.063620	0.041304	-1.037695	0.628520
14	14	0.912931	0.848745	-1.215276	-0.024202
15	15	-0.661489	-1.543994	-0.452754	-1.589521
16	16	-0.272063	-1.012084	0.414147	-0.021171

17	17	0.468146	0.232378	-1.890289	0.827622
18	18	1.744694	0.052282	0.890462	-0.155540
19	19	0.381029	0.666666	0.957832	1.476093
20	20	0.503970	0.567258	1.578919	1.041839
21	21	-0.561937	0.534243	0.246232	0.763021
22	22	0.064446	-0.483406	-0.584869	-1.066767
23	23	2.762885	0.060047	-0.709070	-1.350154
24	24	-0.399912	0.755714	-0.138098	0.771859
25	25	2.289132	-0.178023	0.392030	-1.638884
26	26	-1.667799	-0.723245	-0.298492	-0.464406
27	27	0.871473	-0.875910	1.693428	-0.120278
28	28	2.035892	-0.442804	1.710999	-0.164896
29	29	2.646331	0.006441	-0.819143	-0.140657
30	30	-1.645022	1.841754	-0.390258	-0.118842
31	31	-2.085251	0.043444	-0.183578	0.548298
32	32	1.427635	-0.349981	1.038814	-0.955178
33	33	-0.680210	0.566180	-0.325638	0.244383
34	34	-0.026502	0.758021	0.466427	-0.198146
35	35	0.433309	0.623019	-0.201789	1.983037
36	36	0.646673	-0.715421	-0.322057	0.645001
37	37	0.048681	-1.203145	0.460026	0.192644
38	38	0.098718	-0.672915	-0.229832	-0.187133
39	39	-0.059930	-1.472300	-1.281250	-0.666214
40	40	1.515724	0.207993	-1.037067	-0.393363
41	41	-0.949868	-1.115604	-0.554789	1.324991
42	42	-1.452316	-0.572414	-0.808790	1.615190
43	43	-1.117542	0.083952	-0.874761	0.884389
44	44	0.834118	0.423509	-0.565575	-0.169843
45	45	-0.382312	0.175994	-0.820539	-0.542783

46	46	-0.982393	-0.396464	0.550612	1.584936
47	47	0.581846	-0.994098	0.519327	1.286765
48	48	-0.412282	-1.889874	-0.047348	-0.767669
49	49	0.446292	0.326242	0.230500	0.841655

#to return n rows

df.head(n=5)

Unnamed: 0	A	B	C	D	
0	0	-1.557181	0.373416	1.040685	-0.648513
1	1	-0.835872	-0.005259	0.502099	-0.956839
2	2	-0.257153	2.638228	-0.535731	-1.069443
3	3	-0.505502	-0.339260	0.602622	-1.420995
4	4	1.366398	-0.368296	0.642091	1.609680

 # Return the last n rows.

df.tail(n=6)

	Unnamed: 0	A	B	C	D
44	44	0.834118	0.423509	-0.565575	-0.169843
45	45	-0.382312	0.175994	-0.820539	-0.542783
46	46	-0.982393	-0.396464	0.550612	1.584936
47	47	0.581846	-0.994098	0.519327	1.286765
48	48	-0.412282	-1.889874	-0.047348	-0.767669
49	49	0.446292	0.326242	0.230500	0.841655

 # The index (row labels) of the Dataset.

df.index

RangeIndex(start=0, stop=50, step=1)

#The column labels of the Dataset

df.columns

Index(['Unnamed: 0', 'A', 'B', 'C', 'D'], dtype='object')

#Return a tuple representing the dimensionality of theDataset.

df.shape

(50, 5)

#Return the dtypes in the Dataset

df.dtypes

Unnamed: 0 int64

A float64

B float64

C float64

D float64

dtype: object

#Return the columns values in the Dataset in arrayformat

df.columns.values

array(['Unnamed: 0', 'A', 'B', 'C', 'D'], dtype=object)

#Generate descriptive statistics

df.describe(include='all')

	Unnamed: 0	A	B	C	D
count	50.00000	50.000000	50.000000	50.000000	50.000000
mean	24.50000	0.020103	-0.123273	-0.044716	0.059140
std	14.57738	1.139660	0.845836	0.848863	0.966854
min	0.00000	-2.085251	-2.023668	-1.890289	-1.831875
25%	12.25000	-0.677750	-0.608653	-0.582724	-0.622080
50%	24.50000	-0.090116	0.000591	-0.215810	-0.091596
75%	36.75000	0.578342	0.361622	0.493181	0.813681

max 49.00000 2.762885 2.638228 1.950228 1.983037

#To Read the Data Column wise.

df['A']

0 -1.557181

1 -0.835872

2 -0.257153

3 -0.505502

4 1.366398

5 -1.350101

6 -0.621305

7 -0.670370

8 -1.360577

9 -0.120302

10 0.216936

11 0.567832

12 -0.161148

13 -1.063620

14 0.912931

15 -0.661489

16 -0.272063

17 0.468146

18 1.744694

19 0.381029

20 0.503970

21 -0.561937

22 0.064446

23 2.762885

24 -0.399912

25 2.289132

26   -1.667799

27   0.871473

28   2.035892

29   2.646331

30   -1.645022

31   -2.085251

32   1.427635

33   -0.680210

34   -0.026502

35   0.433309

36   0.646673

37   0.048681

38   0.098718

39   -0.059930

40   1.515724

41   -0.949868

42   -1.452316

43   -1.117542

44   0.834118

45   -0.382312

46   -0.982393

47   0.581846

48   -0.412282

49   0.446292

Name: A, dtype: float64

 # To Sort object by labels (along an axis).

df.sort_index(axis=1,ascending=False)

Unnamed: 0   D   C   B   A

0	0	-0.648513	1.040685	0.373416	-1.557181
1	1	-0.956839	0.502099	-0.005259	-0.835872
2	2	-1.069443	-0.535731	2.638228	-0.257153
3	3	-1.420995	0.602622	-0.339260	-0.505502
4	4	1.609680	0.642091	-0.368296	1.366398
5	5	0.035729	-0.743280	0.210097	-1.350101
6	6	-1.831875	0.141429	-0.620733	-0.621305
7	7	-0.487202	1.950228	-0.379230	-0.670370
8	8	0.998686	-0.576289	0.044675	-1.360577
9	9	1.412786	-0.420956	-0.146491	-0.120302
10	10	0.387263	0.342519	0.398267	0.216936
11	11	-0.064349	-0.315318	-2.023668	0.567832
12	12	-0.932540	-1.226679	-0.118924	-0.161148
13	13	0.628520	-1.037695	0.041304	-1.063620
14	14	-0.024202	-1.215276	0.848745	0.912931
15	15	-1.589521	-0.452754	-1.543994	-0.661489
16	16	-0.021171	0.414147	-1.012084	-0.272063
17	17	0.827622	-1.890289	0.232378	0.468146
18	18	-0.155540	0.890462	0.052282	1.744694
19	19	1.476093	0.957832	0.666666	0.381029
20	20	1.041839	1.578919	0.567258	0.503970
21	21	0.763021	0.246232	0.534243	-0.561937
22	22	-1.066767	-0.584869	-0.483406	0.064446
23	23	-1.350154	-0.709070	0.060047	2.762885
24	24	0.771859	-0.138098	0.755714	-0.399912
25	25	-1.638884	0.392030	-0.178023	2.289132
26	26	-0.464406	-0.298492	-0.723245	-1.667799
27	27	-0.120278	1.693428	-0.875910	0.871473
28	28	-0.164896	1.710999	-0.442804	2.035892

29	29	-0.140657	-0.819143	0.006441	2.646331
30	30	-0.118842	-0.390258	1.841754	-1.645022
31	31	0.548298	-0.183578	0.043444	-2.085251
32	32	-0.955178	1.038814	-0.349981	1.427635
33	33	0.244383	-0.325638	0.566180	-0.680210
34	34	-0.198146	0.466427	0.758021	-0.026502
35	35	1.983037	-0.201789	0.623019	0.433309
36	36	0.645001	-0.322057	-0.715421	0.646673
37	37	0.192644	0.460026	-1.203145	0.048681
38	38	-0.187133	-0.229832	-0.672915	0.098718
39	39	-0.666214	-1.281250	-1.472300	-0.059930
40	40	-0.393363	-1.037067	0.207993	1.515724
41	41	1.324991	-0.554789	-1.115604	-0.949868
42	42	1.615190	-0.808790	-0.572414	-1.452316
43	43	0.884389	-0.874761	0.083952	-1.117542
44	44	-0.169843	-0.565575	0.423509	0.834118
45	45	-0.542783	-0.820539	0.175994	-0.382312
46	46	1.584936	0.550612	-0.396464	-0.982393
47	47	1.286765	0.519327	-0.994098	0.581846
48	48	-0.767669	-0.047348	-1.889874	-0.412282
49	49	0.841655	0.230500	0.326242	0.446292

 # To Sort values by column name.

df.sort_values(by="D")

Unnamed: 0	A	B	C	D	
6	6	-0.621305	-0.620733	0.141429	-1.831875
25	25	2.289132	-0.178023	0.392030	-1.638884
15	15	-0.661489	-1.543994	-0.452754	-1.589521
3	3	-0.505502	-0.339260	0.602622	-1.420995

23	23	2.762885	0.060047	-0.709070	-1.350154
2	2	-0.257153	2.638228	-0.535731	-1.069443
22	22	0.064446	-0.483406	-0.584869	-1.066767
1	1	-0.835872	-0.005259	0.502099	-0.956839
32	32	1.427635	-0.349981	1.038814	-0.955178
12	12	-0.161148	-0.118924	-1.226679	-0.932540
48	48	-0.412282	-1.889874	-0.047348	-0.767669
39	39	-0.059930	-1.472300	-1.281250	-0.666214
0	0	-1.557181	0.373416	1.040685	-0.648513
45	45	-0.382312	0.175994	-0.820539	-0.542783
7	7	-0.670370	-0.379230	1.950228	-0.487202
26	26	-1.667799	-0.723245	-0.298492	-0.464406
40	40	1.515724	0.207993	-1.037067	-0.393363
34	34	-0.026502	0.758021	0.466427	-0.198146
38	38	0.098718	-0.672915	-0.229832	-0.187133
44	44	0.834118	0.423509	-0.565575	-0.169843
28	28	2.035892	-0.442804	1.710999	-0.164896
18	18	1.744694	0.052282	0.890462	-0.155540
29	29	2.646331	0.006441	-0.819143	-0.140657
27	27	0.871473	-0.875910	1.693428	-0.120278
30	30	-1.645022	1.841754	-0.390258	-0.118842
11	11	0.567832	-2.023668	-0.315318	-0.064349
14	14	0.912931	0.848745	-1.215276	-0.024202
16	16	-0.272063	-1.012084	0.414147	-0.021171
5	5	-1.350101	0.210097	-0.743280	0.035729
37	37	0.048681	-1.203145	0.460026	0.192644
33	33	-0.680210	0.566180	-0.325638	0.244383
10	10	0.216936	0.398267	0.342519	0.387263
31	31	-2.085251	0.043444	-0.183578	0.548298

13	13	-1.063620	0.041304	-1.037695	0.628520
36	36	0.646673	-0.715421	-0.322057	0.645001
21	21	-0.561937	0.534243	0.246232	0.763021
24	24	-0.399912	0.755714	-0.138098	0.771859
17	17	0.468146	0.232378	-1.890289	0.827622
49	49	0.446292	0.326242	0.230500	0.841655
43	43	-1.117542	0.083952	-0.874761	0.884389
8	8	-1.360577	0.044675	-0.576289	0.998686
20	20	0.503970	0.567258	1.578919	1.041839
47	47	0.581846	-0.994098	0.519327	1.286765
41	41	-0.949868	-1.115604	-0.554789	1.324991
9	9	-0.120302	-0.146491	-0.420956	1.412786
19	19	0.381029	0.666666	0.957832	1.476093
46	46	-0.982393	-0.396464	0.550612	1.584936
4	4	1.366398	-0.368296	0.642091	1.609680
42	42	-1.452316	-0.572414	-0.808790	1.615190
35	35	0.433309	0.623019	-0.201789	1.983037

#To generate Pure integer-location based indexing for selection byposition.

df.iloc[5]

```
Unnamed: 0    5.000000
A            -1.350101
B             0.210097
C            -0.743280
D             0.035729
Name: 5, dtype: float64
```

 # Selecting via [], which slices the rows.

df[0:3]

Unnamed: 0	A	B	C	D	
0	0	-1.557181	0.373416	1.040685	-0.648513
1	1	-0.835872	-0.005259	0.502099	-0.956839
2	2	-0.257153	2.638228	-0.535731	-1.069443

#Selection by label

df.loc[:,["A","B"]]

A	B	
0	-1.557181	0.373416
1	-0.835872	-0.005259
2	-0.257153	2.638228
3	-0.505502	-0.339260
4	1.366398	-0.368296
5	-1.350101	0.210097
6	-0.621305	-0.620733
7	-0.670370	-0.379230
8	-1.360577	0.044675
9	-0.120302	-0.146491
10	0.216936	0.398267
11	0.567832	-2.023668
12	-0.161148	-0.118924
13	-1.063620	0.041304
14	0.912931	0.848745
15	-0.661489	-1.543994
16	-0.272063	-1.012084
17	0.468146	0.232378
18	1.744694	0.052282
19	0.381029	0.666666
20	0.503970	0.567258

21	-0.561937	0.534243
22	0.064446	-0.483406
23	2.762885	0.060047
24	-0.399912	0.755714
25	2.289132	-0.178023
26	-1.667799	-0.723245
27	0.871473	-0.875910
28	2.035892	-0.442804
29	2.646331	0.006441
30	-1.645022	1.841754
31	-2.085251	0.043444
32	1.427635	-0.349981
33	-0.680210	0.566180
34	-0.026502	0.758021
35	0.433309	0.623019
36	0.646673	-0.715421
37	0.048681	-1.203145
38	0.098718	-0.672915
39	-0.059930	-1.472300
40	1.515724	0.207993
41	-0.949868	-1.115604
42	-1.452316	-0.572414
43	-1.117542	0.083952
44	0.834118	0.423509
45	-0.382312	0.175994
46	-0.982393	-0.396464
47	0.581846	-0.994098
48	-0.412282	-1.889874
49	0.446292	0.326242

#To print subset of the first n columns of the original data

df.iloc[:,:4]

	Unnamed: 0	A	B	C
0	0	-1.557181	0.373416	1.040685
1	1	-0.835872	-0.005259	0.502099
2	2	-0.257153	2.638228	-0.535731
3	3	-0.505502	-0.339260	0.602622
4	4	1.366398	-0.368296	0.642091
5	5	-1.350101	0.210097	-0.743280
6	6	-0.621305	-0.620733	0.141429
7	7	-0.670370	-0.379230	1.950228
8	8	-1.360577	0.044675	-0.576289
9	9	-0.120302	-0.146491	-0.420956
10	10	0.216936	0.398267	0.342519
11	11	0.567832	-2.023668	-0.315318
12	12	-0.161148	-0.118924	-1.226679
13	13	-1.063620	0.041304	-1.037695
14	14	0.912931	0.848745	-1.215276
15	15	-0.661489	-1.543994	-0.452754
16	16	-0.272063	-1.012084	0.414147
17	17	0.468146	0.232378	-1.890289
18	18	1.744694	0.052282	0.890462
19	19	0.381029	0.666666	0.957832
20	20	0.503970	0.567258	1.578919
21	21	-0.561937	0.534243	0.246232
22	22	0.064446	-0.483406	-0.584869
23	23	2.762885	0.060047	-0.709070
24	24	-0.399912	0.755714	-0.138098

25	25	2.289132	-0.178023	0.392030
26	26	-1.667799	-0.723245	-0.298492
27	27	0.871473	-0.875910	1.693428
28	28	2.035892	-0.442804	1.710999
29	29	2.646331	0.006441	-0.819143
30	30	-1.645022	1.841754	-0.390258
31	31	-2.085251	0.043444	-0.183578
32	32	1.427635	-0.349981	1.038814
33	33	-0.680210	0.566180	-0.325638
34	34	-0.026502	0.758021	0.466427
35	35	0.433309	0.623019	-0.201789
36	36	0.646673	-0.715421	-0.322057
37	37	0.048681	-1.203145	0.460026
38	38	0.098718	-0.672915	-0.229832
39	39	-0.059930	-1.472300	-1.281250
40	40	1.515724	0.207993	-1.037067
41	41	-0.949868	-1.115604	-0.554789
42	42	-1.452316	-0.572414	-0.808790
43	43	-1.117542	0.083952	-0.874761
44	44	0.834118	0.423509	-0.565575
45	45	-0.382312	0.175994	-0.820539
46	46	-0.982393	-0.396464	0.550612
47	47	0.581846	-0.994098	0.519327
48	48	-0.412282	-1.889874	-0.047348
49	49	0.446292	0.326242	0.230500

 # To Slice the data

df.iloc[3:5,0:2]

Unnamed: 0 A

3	3	-0.505502
4	4	1.366398

#to check missing values or null values

df.isnull()

Unnamed: 0	A	B	C	D
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
5	False	False	False	False	False
6	False	False	False	False	False
7	False	False	False	False	False
8	False	False	False	False	False
9	False	False	False	False	False
10	False	False	False	False	False
11	False	False	False	False	False
12	False	False	False	False	False
13	False	False	False	False	False
14	False	False	False	False	False
15	False	False	False	False	False
16	False	False	False	False	False
17	False	False	False	False	False
18	False	False	False	False	False
19	False	False	False	False	False
20	False	False	False	False	False
21	False	False	False	False	False
22	False	False	False	False	False
23	False	False	False	False	False

24	False	False	False	False	False
25	False	False	False	False	False
26	False	False	False	False	False
27	False	False	False	False	False
28	False	False	False	False	False
29	False	False	False	False	False
30	False	False	False	False	False
31	False	False	False	False	False
32	False	False	False	False	False
33	False	False	False	False	False
34	False	False	False	False	False
35	False	False	False	False	False
36	False	False	False	False	False
37	False	False	False	False	False
38	False	False	False	False	False
39	False	False	False	False	False
40	False	False	False	False	False
41	False	False	False	False	False
42	False	False	False	False	False
43	False	False	False	False	False
44	False	False	False	False	False
45	False	False	False	False	False
46	False	False	False	False	False
47	False	False	False	False	False
48	False	False	False	False	False
49	False	False	False	False	False

df.notnull

<bound method DataFrame.notnull of   Unnamed: 0   A   B   C   D

0	0	-1.557181	0.373416	1.040685	-0.648513
1	1	-0.835872	-0.005259	0.502099	-0.956839
2	2	-0.257153	2.638228	-0.535731	-1.069443
3	3	-0.505502	-0.339260	0.602622	-1.420995
4	4	1.366398	-0.368296	0.642091	1.609680
5	5	-1.350101	0.210097	-0.743280	0.035729
6	6	-0.621305	-0.620733	0.141429	-1.831875
7	7	-0.670370	-0.379230	1.950228	-0.487202
8	8	-1.360577	0.044675	-0.576289	0.998686
9	9	-0.120302	-0.146491	-0.420956	1.412786
10	10	0.216936	0.398267	0.342519	0.387263
11	11	0.567832	-2.023668	-0.315318	-0.064349
12	12	-0.161148	-0.118924	-1.226679	-0.932540
13	13	-1.063620	0.041304	-1.037695	0.628520
14	14	0.912931	0.848745	-1.215276	-0.024202
15	15	-0.661489	-1.543994	-0.452754	-1.589521
16	16	-0.272063	-1.012084	0.414147	-0.021171
17	17	0.468146	0.232378	-1.890289	0.827622
18	18	1.744694	0.052282	0.890462	-0.155540
19	19	0.381029	0.666666	0.957832	1.476093
20	20	0.503970	0.567258	1.578919	1.041839
21	21	-0.561937	0.534243	0.246232	0.763021
22	22	0.064446	-0.483406	-0.584869	-1.066767
23	23	2.762885	0.060047	-0.709070	-1.350154
24	24	-0.399912	0.755714	-0.138098	0.771859
25	25	2.289132	-0.178023	0.392030	-1.638884
26	26	-1.667799	-0.723245	-0.298492	-0.464406
27	27	0.871473	-0.875910	1.693428	-0.120278
28	28	2.035892	-0.442804	1.710999	-0.164896

```
29        29  2.646331  0.006441 -0.819143 -0.140657

30        30 -1.645022  1.841754 -0.390258 -0.118842

31        31 -2.085251  0.043444 -0.183578  0.548298

32        32  1.427635 -0.349981  1.038814 -0.955178

33        33 -0.680210  0.566180 -0.325638  0.244383

34        34 -0.026502  0.758021  0.466427 -0.198146

35        35  0.433309  0.623019 -0.201789  1.983037

36        36  0.646673 -0.715421 -0.322057  0.645001

37        37  0.048681 -1.203145  0.460026  0.192644

38        38  0.098718 -0.672915 -0.229832 -0.187133

39        39 -0.059930 -1.472300 -1.281250 -0.666214

40        40  1.515724  0.207993 -1.037067 -0.393363

41        41 -0.949868 -1.115604 -0.554789  1.324991

42        42 -1.452316 -0.572414 -0.808790  1.615190

43        43 -1.117542  0.083952 -0.874761  0.884389

44        44  0.834118  0.423509 -0.565575 -0.169843

45        45 -0.382312  0.175994 -0.820539 -0.542783

46        46 -0.982393 -0.396464  0.550612  1.584936

47        47  0.581846 -0.994098  0.519327  1.286765

48        48 -0.412282 -1.889874 -0.047348 -0.767669

49        49  0.446292  0.326242  0.230500  0.841655>
```

#Import Library

from sklearn import preprocessing

df=pd.read_csv(r"C:\Users\System21\Downloads\iris.csv")

df.head()

	sepal.length	sepal.width	petal.length	petal.width	type
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa

2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa